Practice Final

Math 3150 001 Summer 2015
C.Sampson
June 24th 2014 12:30pm-2:00pm in our class room **Name and Unid:**

by writing my name I swear by the honor code

Read all of the following information before starting the exam:
Note this is the practice exam ignore point values on problems there are extras but the exam will look similar

- Show all work, clearly and in order, if you want to get full credit. I reserve the right to take off points if I cannot see how you arrived at your answer (even if your final answer is correct).
- Justify your answers algebraically whenever possible to ensure full credit. Sketch all relevant graphs and explain all relevant mathematics.
- Circle or otherwise indicate your final answers.
- You MAY NOT use a calculator or notes. You may use the formulas provided and your 3X5 card.
- The actual test has 8 problems and is worth 150 points It is your responsibility to make sure that you have all of the pages!
- Good luck!

Good Luck!
Here are some Formulas and rules for you.

1. (*10 points*) Find the equilibrium temperature distribution of a rod of length $L = 1$ with no internal heat source and insulated boundaries. That is, subject to the boundary conditions $\frac{\partial u}{\partial x}(0) = 0$ and $\frac{\partial u}{\partial x}(1) = 0$, with an initial temperature distribution of $u(x, 0) = 3x^2$

2. (*10 points*) Prove that the total thermal energy is constant for a 1-d rod of length L when it has perfectly insulated boundaries and no internal sources.

3. (*10 points*) For Laplace's equation in 2-d cartesian coordinates, $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ apply the assumption of separation of variables $u(x, y) = h(x)\phi(y)$ and derive the resulting ODE equations.

4. (*10 points*) For Laplace's equation in polar coordinates, $\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$ apply the assumption of separation of variables $u(r,\theta) = G(r)\phi(\theta)$ and derive the resulting ODE equations.

5. (*10 points*) For the 1-d wave equation $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$ apply the assumption of separation of variables, $u(x,t) = \phi(x)h(t)$ and derive the resulting ODE equations.

6. (*10 points*) Solve Laplace's equation inside a rectangle $0 \leq x \leq L$, $0 \leq y \leq H$ with the following boundary conditions. $\frac{\partial u}{\partial y}(x,0) = 0$, $\frac{\partial u}{\partial y}(x,H) = 0$, $u(0,y) = 0$, $u(L,y) = f(y)$

7. (*20 points*) Consider the general solution Laplace's equation inside a circle of radius $r = a$ given by

$$u(r,\theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n cos(n\theta) + B_n r^n sin(n\theta)$$

a. (*10 pts*) Find the full solution if on the boundary, $u(a,\theta) = 1 + 2\sin(3\theta) + 12\sin(\theta) + 3\cos(4\theta)$

b. (*10 pts*) Find the full solution if on the boundary, $u(a,\theta) = 1$

8. (*10 points*) Consider the two ODE's coming from Laplace's equation in polar coordinates, $\frac{d^2\phi}{d\theta^2} = -\lambda\phi$ and $\frac{r}{G}\frac{d}{dr}\left(r\frac{dG}{dr}\right) = \lambda$

a. (*5 pts*) Find the general solution to both if $\lambda = 0$

b. (*5 pts*) Find the general solution to both if $\lambda > 0$

9. (*15 points*) For the following functions sketch $f(x)$, the Fourier series for $f(x)$, the Fourier Sine series of $f(x)$, and the Fourier Cosine series of $f(x)$ on the interval $-L < x < L$.

a. (*5 pts*) $f(x) = x$

b. (*5 pts*) $f(x) = -x(x - L)$

c. (*5 pts*) $f(x) = 1$ for $x > 0$ and $f(x) = 2$ for $x < 0$

10. (*5 points*)
Find the Fourier Series coefficients for $f(x) = x$ on the interval $-L < x < L$

11. (*5 points*) For the following functions state whether it has a continuous Fourier cosine series or continuous Fourier sine series neither or both.

a. (*1 pt*) $f(x) = 1$
b. (*1 pt*) $f(x) = x$
c. (*1 pt*) $f(x) = -(x)(x - L)$
d. (*1 pt*) $f(x) = 1 + x$

12. (*10 points*) Verify that $u(x,t) = \int_0^\infty e^{-k\lambda t}\left(c_1\cos(\sqrt{\lambda}x) + c_2\sin(\sqrt{\lambda}x)\right)d\lambda$ is a solution to the 1-d heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$

13. (*10 points*) Solve the 1-d wave equation $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$ on the whole real line subject to the initial conditions $u(x,0) = f(x)$ and $\frac{\partial u}{\partial t}(x,0) = 0$

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